Bence Jordan Danko

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EDUCATION

San Jose State University — 2024 – Present
M.Sc. in Applied Data Intelligence — San Jose, USA

San Jose State University — 2020 – 2024
B.Sc. in Computer Science — San Jose, USA

PROJECTS

Segmentation/OCR Multimodal API Agent for Grazen AI

https://warehouse.openwear.ai

- Collaborated with startup researchers to develop a multimodal API agent for warehouse logistics to automate inventory audits.
- Utilizes SAM 3 for object localization and GLM-OCR for text extraction to reduce manual labor and inventory losses.
- Implemented an orchestration pipeline that integrates SAM 3 segmentation with LLM-based filtering, allowing for dynamic re-prompting and precise localization of front-row product stacks.
- Built a comprehensive data extraction service that combines bounding box analysis and OCR processing to synthesize structured inventory reports, including product names, instance counts, and quantity validation.
- Engineered a dashboard for displaying real-time segmented crops, detected product counts, and extracted textual labels for warehouse management validation.

Edge Vision Smart Elevators: Privacy-Preserving Headcount Analysis

https://github.com/bencejdanko/m55m1-ElevatorCounting-YOLOv8n

- Conducting applied research for Nuvoton's NuMicro M55M1 microcontroller, architecting a confined person-recognition pipeline achieving 15 FPS within 1.5MB SRAM constraints.
- Engineered an INT8-quantized YOLOv8n pipeline using the Vera compiler and Arm Ethos-U55 NPU.
- Curated a specialized dataset of 13,000+ images from Roboflow, and distributed for common public use on Huggingface.
- Assembled a proprietary data collection setup using a Raspberry Pi Zero W and 160ř FOV camera for real-world confined elevator environments.
- Developed custom firmware using CMSIS-CSolution to satisfy business presentation requirements.
- Collaborated with Nuvoton hardware engineers to validate system performance for a live demonstration at the Sensors Converge conference.

Mobilint MLA100 NPU Computer Vision Applications

https://github.com/bencejdanko/imagenet-classification-mobilint-mla-100

- Worked with Mobilint hardware and compiler engineers to evaluate computer vision applications for the MLA100 NPU.
- Developed and optimized a real-time image restoration pipeline for the NPU, achieving <6ms latencies for 256x256 images.
- Replaced a high-latency ResNet baseline (400ms+) with a custom 0.2M parameter Convolutional Network by optimizing kernel launches and restricting feature maps to 64 to minimize hardware-induced latency.
- Orchestrated large-scale training and data processing jobs using modal, scaling cloud GPU resources to conduct progressive training cycles with Cosine Annealing and mixed precision.
- Optimized PSNR improvement through iterative checkpoint-based training and rigorous hardware-aware model profiling.
- Evaluated and compiled ImageNet classification models for the MLA100, comparing inference throughput against Intel AI Boost and achieving 2.7x performance gains.